

# The Geometry of the Padding-Bottom Hack

Exploiting CSS Specifications for Intrinsic Aspect Ratios

## 📖 CSS Box Model Spec

Why does `padding-bottom` create height relative to width?

Because the W3C CSS Spec explicitly states:

*"The percentage is calculated with respect to the **width** of the generated box's containing block..."*

This quirk applies to **both** `padding-top` and `padding-bottom`!

## 🧮 The 16:9 Math

We need an empty box where the Height is exactly proportional to the Width.

$$\text{Ratio} = \frac{\text{Height}}{\text{Width}}$$

$$16 : 9 \rightarrow \frac{9}{16} = 0.5625$$

Expressed as a percentage of the width:

$$0.5625 \times 100\% = 56.25\%$$

